

Claudio Mazzi

Ph.D. Candidate in Artificial Intelligence for Society | Research Collaborator

Department of Computer Science, University of Pisa | L'EMbeDS & MeS Laboratory, Sant'Anna School of Advanced Studies, Pisa, Italy

✉ claudio.mazzi@santannapisa.it • 📍 Pisa, Italy • 🆔 ORCID: [0009-0007-9407-9897]
🐙 GitHub • 🔗 LinkedIn

Research Profile & Core Methodological Expertise

Physicist by training and doctoral candidate in Artificial Intelligence for Society. Research focuses on the development and application of machine learning, explainable artificial intelligence (XAI), and computational statistics to solve critical challenges in healthcare systems.

- **Machine Learning & Explainable AI in Medicine:** Predictive risk stratification, disease complication forecasting in chronic populations (Type-2 Diabetes), NLP-driven classification of clinical risk and workplace violence in healthcare services, and model interpretability.
- **Actuarial Statistics & Stochastic Reserving:** Stochastic loss-reserving methodologies for medical malpractice liabilities and claims estimation in public healthcare systems.
- **Functional Data Analysis:** Synthetic functional data generation for data augmentation and privacy preserving systems.
- **Health Systems Performance Benchmarking:** Data pipeline engineering, statistical auditing, and quality benchmarking across administrative health records.

Education

Ph.D. in Artificial Intelligence for Society

University of Pisa & Sant'Anna School of Advanced Studies

Nov 2023 – Oct 2026 (Expected)

Pisa, Italy

- *National Ph.D. Program in Artificial Intelligence (39th Cycle)*, funded by the Italian Ministry of University and Research (MUR). Joint degree between University of Pisa and Sant'Anna School of Advanced Studies.
- *Institutional Affiliations:* Department of Computer Science (UniPi); L'EMbeDS (Economics, Management and Data Science for Society) and MeS Laboratory, Sant'Anna School of Advanced Studies.
- *Doctoral Research Focus:* Advanced statistical machine learning, explainable AI and NLP methods for healthcare decision support and health analytics. Advisor: Prof. Chiara Seghieri.

Master's Degree in Physics

University of Milan-Bicocca

2017 – 2019

Milan, Italy | Final Grade: 108 / 110

- *Dissertation:* Large Charge Expansion and Semiclassical Method. The XYZ-model in BPS and superfluid phase.

Bachelor's Degree in Physics

University of Milan-Bicocca

2013 – 2016

Milan, Italy | Final Grade: 98 / 110

Visiting Research Appointments

Visiting Researcher

Department of Mathematics and Statistics, Auburn University

Sep 2026 – Oct 2026

Auburn, Alabama, USA

- *Host & Academic Collaborator:* Prof. Roberto Molinari.
- *Research Program:* Methodological research in functional data analysis (FDA) and non-stationary kernel engines. Investigating RKHS inductive bias and orthonormal splinet expansions for synthetic functional data generation from single longitudinal trajectories ($N = 1$).

Academic & Research Appointments

Research Collaborator (*Collaboratore di Ricerca - Co.co.co.*)

Jan 2024 – Present

MeS Laboratory, Sant'Anna School of Advanced Studies

Pisa, Italy

- **Healthcare Registry Architecture & Governance:** Lead data manager responsible for the architectural design, ingestion pipelines, automated ETL workflows, and statistical quality assurance of regional administrative healthcare databases, specifically:
 - Specialized Outpatient Care Information System (*Prestazioni Ambulatoriali - SPA*).
 - Emergency Department Admissions and Clinical Records (*Pronto Soccorso - PS*).
- **Performance Evaluation System (MeS System):** Direct the computational execution, metric validation, and cross-regional benchmarking of over **140 performance indicators** within the Inter-Regional Healthcare Performance Evaluation System. Domains encompass emergency care efficiency, outpatient care access, institutional scientific productivity, and clinical risk.
- **Scientific Lead for Healthcare Claims Reserving:** Principal investigator and lead methodologist for the actuarial and statistical estimation of healthcare claims reserves (*riserva sinistri*) for the Tuscany Regional Health Authority (*Regione Toscana*); engineered and deployed an advanced stochastic claims reserving model formally integrated into regional administrative decision-making.

Research Fellow

Mar 2025 – Aug 2025

Department of Cultures and Societies, University of Palermo

Palermo, Italy

- **National Research Project PRIN RETIN:** Acted as research data scientist analyzing large-scale Chronic Related Group (*CReG*) longitudinal registries for cohorts of patients diagnosed with Type-2 Diabetes Mellitus (T2DM).
- **Predictive Analytics & Risk Stratification:** Developed and validated machine learning and explainable AI (XAI) models to predict the onset and progression of micro- and macro-vascular complications in diabetic patients.
- **Cross-Regional Epidemiological Study:** Coordinated data linkage and statistical modeling across a joint cohort comprising data from Sicily, Tuscany and Lombardy, establishing comparative benchmarking metrics.

Research Fellow

Dec 2022 – Oct 2023

MeS Laboratory, Sant'Anna School of Advanced Studies

Pisa, Italy

- **Interdisciplinary Actuarial Research:** Conducted collaborative research with The BioRobotics Institute on stochastic loss-reserving techniques for medical malpractice liability in public hospitals; the developed mathematical framework was officially adopted and operationalized by the Tuscany Regional Health Authority.
- **Data Infrastructure Management:** Managed the administrative data pipelines for outpatient (SPA) and emergency (PS) registries.
- **Healthcare Workforce Well-Being Project:** Co-designed statistical pipelines and authored two comprehensive evaluation monographs on organizational climate and healthcare personnel well-being for the Tuscany Region.

Research Collaborator (*Collaboratore di Ricerca - Co.co.co.*)

Jul 2022 – Nov 2022

MeS Laboratory, Sant'Anna School of Advanced Studies

Pisa, Italy

- Supervised data cleaning, consolidation, and indicator computation for outpatient services, emergency records, and regional health research monitoring.

High School Teacher of Mathematics and Physics

Sep 2020 – Jun 2022

Liceo Melchiorre Gioia & Liceo Scientifico Lorenzo Respighi

Piacenza, Italy

Publications & Scholarly Output

Articles in Peer-Reviewed International Journals

1. **Claudio Mazzi**, Angelo Damone, Andrea Vandelli, Gastone Ciuti, and Milena Vainieri (2024). “Stochastic Claims Reserve in the Healthcare System: A Methodology Applied to Italian Data”. *Risks*, 12(2), 24.
 DOI: 10.3390/risks12020024

Refereed Conference Proceedings & Book Chapters

2. **C. Mazzi**, Roberto Molinari, and Chiara Seghieri (2025). “Machine Learning and Explainable AI for Type-2 Diabetes Management”. In: *Statistical Learning, Data Science and Complex Data*, pp. 175–180. Springer Nature Switzerland.
 DOI: 10.1007/978-3-031-96736-8_30
3. Chiara Ferrara, Matilde Bisi, **Claudio Mazzi**, Francesca Chiaromonte, Paolo Ferragina, Massimo Ughi, and Chiara Seghieri (2026). “Identifying Structural Typologies of Workplace Aggression in Healthcare: A Text-Validated Analysis of Toscana Nord Ovest Reports”. In: *Statistical Science: From Theory to Applied Research IV*, Springer Nature Switzerland, Cham.
 DOI: 10.1007/978-3-032-30665-4_7
 **Winner of the Best Poster Award at the SIS FENStatS 2026 Conference.**

Working Papers & Preprints

4. **Claudio Mazzi**, Roberto Molinari, and Jacopo Di Iorio (2026). “Synthetic Functional Data Generation from a Single Trajectory ($N = 1$): RKHS Inductive Bias, Orthonormal Splinets, and the Splinet-Modulated Non-Stationary (SMNS) Kernel Engine”. *Working Paper / In Preparation for Submission*.

Manuscripts Under Peer Review

5. Benedetta Dal Canto, **Claudio Mazzi**, Giaeale Moretti, Bruna Vinci, Domenico Di Giorgio, Oscar Cruciani, Stefania Vasselli, and Milena Vainieri (2026). “Predicting Drug Shortages: An Explainable Machine Learning Study Using Administrative Data”. Submitted to: *Health Care Management Science*.
6. **Claudio Mazzi**, Chiara Seghieri, Roberto Molinari, Paolo Berta, and Mauro Ferrante (2026). “Machine Learning and Explainable AI for Type-2 Diabetes Management: A Multi-Regional Study Using Administrative Health Data”. Submitted to: *Journal of the Royal Statistical Society: Series C*.
7. **Claudio Mazzi**, Matilde Contestabile, Paolo Ferragina, Monica Chillà, Gaetana Morgante, Francesca Chiaromonte, Massimo Ughi, and Chiara Seghieri (2026). “Locally Deployable AI for Legal Classification and Risk Stratification of Workplace Violence in Healthcare: A Retrospective Study from Italy”. Available at SSRN:
 SSRN: 7165564 | DOI: 10.2139/ssrn.7165564

Institutional Research Monographs & Policy Evaluation Reports

8. Nicola Bellè, Paola Cantarelli, Domenico Cerasuolo, Giuseppe D’Orio, **Claudio Mazzi**, and Luca Scopis (2023). *Indagine di clima organizzativo 2023 - Azienda Sanitaria dell’Alto Adige*. Sant’Anna School of Advanced Studies.
 Handle: 11382/561352
9. Nicola Bellè, Paola Cantarelli, Luca Pirrotta, Luca Scopis, Veronica Cruciani, Domenico Cerasuolo, **Claudio Mazzi**, and Giuseppe D’Orio (2023). *Indagine di clima organizzativo - Sistema sanitario della Toscana - 2023*. Sant’Anna School of Advanced Studies.
 Handle: 11382/555572
10. Nicola Bellè, Paola Cantarelli, Luca Pirrotta, Luca Scopis, **Claudio Mazzi**, and Giuseppe D’Orio (2023). *Indagine di clima organizzativo - Sistema socio sanitario del Veneto - 2023*. Sant’Anna School of Advanced Studies.
 Handle: 11382/553332

Conference Presentations & Scientific Dissemination

Oral Presentations at National & International Conferences

- **SIS 2025 Statistics for Innovation** — Intermediate Scientific Meeting of the Italian Statistical Society, 2025. 16-18 June 2025, University of Genova, Genova, Italy.
- **L'EMbeDS and FAIR SMaRT COnSTRUCT** — Learning from large, complex and structured data: advances in methods and applications. 3 June 2025. Sant'Anna School of Advanced Studies, Pisa, Italy.

Peer-Reviewed Conference Abstracts & Posters

- **SIS-FENStatS 2026** — 52nd Scientific Meeting of the Italian Statistical Society, 2026. 22-25 June, University La Sapienza, Roma, Italy.
- **CS2Italy 2026** — The Italian Conference On Computational Social Science. 19-21 May 2026. OGR, Turin, Italy.
- **Claudio Mazzi**, Chiara Seghieri, Emilio Incerti, Francesca Grieco, and Alberto Piaggese (2026). “Uncovering Heterogeneity in Diabetic Foot Care Pathways through Stochastic Conformance Checking”. *Diabetic Foot Study Group (DFSG) Annual Meeting*, Florence, Italy, September 2026.

Teaching & Professional Training Experience

Teaching Assistant

A.Y. 2024/25 & 2025/26

Sant'Anna School of Advanced Studies

Pisa, Italy

- **Course: Applied Statistical Modeling** (Lead Instructor: Prof. Chiara Seghieri).
- Taught methodological and applied foundations of statistical models for analyzing diverse response variables and complex data structures. Covered theoretical properties, parameter estimation methods, and practical applications of linear regression, Generalized Linear Models (GLMs), and an introduction to multivariate data analysis.
- Designed computational lab problem sets and led hands-on programming tutorials in **R**.

Executive Education Instructor & Project Work Advisor

A.Y. 2024/2025

Sant'Anna School of Advanced Studies (Executive Healthcare Leaders Program)

Pisa, Italy

- Faculty advisor for executive capstone projects presented by hospital chief medical officers and directors of complex healthcare operational units (*Dirigenti di Struttura Complessa*).
- **Domain Focus:** Emergency Department operations management, patient throughput modeling, bottleneck analytics, and triage overcrowding alleviation.

Advanced Courses & Specialized Training

- **Doctoral School on Artificial Intelligence and Cybersecurity**, Sapienza University of Rome, Italy **4–6 Dec 2023**
- **ECDL / ICDL (European Computer Driving Licence)**, AICA **Jun 2010**

Skills & Competencies

- **Languages:** Italian (Native); English (**C1 Proficient** — CEFR Certificate, score 97/100, Excellent); French (**A1** — Elementary).
- **Programming & Scientific Computing:** Python, R, SAS, C/C++, MATLAB, Mathematica.
- **Typesetting & Tools:** L^AT_EX, Git, GitHub, Markdown, Microsoft Office Suite, Power BI, Adobe InDesign, Canva.

In compliance with the Italian Legislative Decree no. 196/2003 and the GDPR (EU 2016/679), I authorize the processing of my personal data for recruitment purposes.